PRE-OPERATION INSPECTION

- Ensure that the mill has adequate water supply to begin operations.
- Check pump pressure to ensure that water pressure is set to manufacturer's recommended specifications.
- Check individual spray bar locations to ensure that all nozzles are functioning correctly with a good spray pattern.
- Verify that all nozzles are functioning properly and spray pattern is correct for the designated location.



WARNING

TO AVOID INJURY:

- **Read and understand manuals before operating.**
- **Keep hands clear of moving conveyor.**
- **Keep crew clear during machine movement.**
- **Stay away from cutter and discharge areas.**

PREPARE FOR SAFE OPERATION

Before you begin to operate the machine, take time to check that it is in good working condition.

- Check and use all available protective and safety devices, such as railings, safety tread, hand-holds, and interlock devices.
- Perform daily and periodic service procedures as instructed by the equipment manufacturers.
- Check for broken, missing or damaged parts and loose or missing fasteners. Make any necessary repairs.
- Check that no warning tags have been placed on the machine.
- Check that warning signs, special instructions, and operator's manuals are readable and in the proper location.
- Ensure all doors, safety devices, guards, and covers are in place and secured properly.

Operator training

- The operator must demonstrate a thorough understanding of the dust suppression systems on the machine:
  - ◊ Water system operation
    - ◆ Water application locations within the machine
    - ◆ Flow rates
    - ◆ Spray patterns
    - ◆ Anticipated water usage per unit time or per unit square yard.
    - ◆ Maintenance - ensure correct nozzles per manufacturer's specifications.
    - ◆ Troubleshooting
    - ◆ Check in-line water filter
  - ◊ Visual inspection of seals, flashing, and enclosures, to ensure minimal dust leakage.

OPERATION

- Maintain clean water supply to the milling machine. Maintaining a clean water supply will reduce the chances of clogging strainers and nozzles.
- Monitor clean nozzles and water switch or flow control settings and system pressure to ensure proper application rate.
- Monitor conveyor belt speeds. Slow conveyer speed can cause material to stay in the housing longer, thereby reducing production and possible creating more airborne dust.
- Inspect and replace cutting tools (bits/teeth) as needed. Dull teeth reduce cutting efficiency, thereby reducing production and possibly creating more airborne dust.
- Front moldboard setting: lower front moldboard and / or primary conveyor to maximize housing exit window area. Reducing the exit window area will keep material in the housing longer, reducing production and creating more airborne dust.
- Make sure that the cutter drum kicker paddles are in good condition. Worn-out paddles will allow material to stay in the drum housing longer, reducing production and creating more airborne dust.
- Inspect the flashing that seals the conveyor frame to the belt. If worn and not replaced, dust / material exits the conveyor prematurely and can cause significant dust issues.

Appendix G: Glossary

Asphalt pavement milling machine: A milling machine or cold-planer is a type of highway construction machinery equipped with a toothed rotary cutter drum and conveyor system capable of rapid removal of recycled asphalt pavement.

Cutter drum: A rotary drum equipped with cutting teeth for removing recycled asphalt pavement.

Drum housing: The area inside the enclosure around the rotary cutter drum.

Ground person: The worker whose primary function is to operate the controls mounted to the sides of the machine at ground level.

Half-lane and larger milling machine: A half-lane cold planer that typically weighs more than 60,000 pounds and has a cutter drum width greater than 4 feet.

Operator: The worker whose primary function is to control the milling machine's speed, direction, and conveyors.

Primary conveyor: The initial lift conveyor that transports recycled asphalt pavement from the cutter drum housing to the secondary conveyor.

Respirable crystalline silica: "That portion of airborne crystalline silica that is capable of entering the gas-exchange regions of the lungs if inhaled; by convention, a particle-size-selective fraction of the total airborne dust; includes particles with aerodynamic diameters less than approximately 10 μm and has a 50% deposition efficiency for particles with an aerodynamic diameter of approximately 4 μm" [NIOSH 2002].

Secondary conveyor: The longer conveyor that transports recycled asphalt pavement from the primary lift conveyor to a dump truck.

Reference

NIOSH [2002]. NIOSH hazard review: health effects of occupational exposure to respirable crystalline silica. Cincinnati, OH: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, National Institute for Occupational Safety and Health, DHHS (NIOSH) Publication No. 2002-129.



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